

experiment report

Below is a publication-ready experiment report template tailored to your system. It is structured so you can both defend the methodology and demonstrate measurable gains across phases. Avoid padding—reviewers care about clarity, controls, and reproducibility.

experiment_report.md

Title

A Multi-Stage Trust, Network, and Spatially-Aware Framework for Robust Crowd-Based Credibility Estimation

Abstract

We evaluate a staged system that estimates content credibility using user behavior, network structure, spatial context, and optional machine learning augmentation. Using a controlled simulation framework, we quantify performance under benign conditions and adversarial scenarios. Results show progressive improvements in robustness, early detection, and alert precision across system phases, with the largest gains arising from coordination-aware graph modeling and spatial filtering.

1. Problem Definition

Crowd-sourced systems are vulnerable to:

- noisy participation
- coordinated manipulation
- lack of contextual relevance

We aim to estimate:

$C_j(t)$

such that:

- it converges to ground truth y_j
- it resists adversarial behavior
- it enables reliable alerting

2. System Variants (Ablation Setup)

You must not evaluate only the final system. Compare incremental phases:

Model Variants

M0: Baseline (unweighted voting) M1: MVP (Bayesian + time decay) M2: + User-aware (R_i , Exp_i , $Anom_i$)
M3: + Graph trust (T_i , coordination) M4: + Spatial (Prox, L_i) M5: + ML augmentation (C_{ML} , C_{memory})

3. Dataset (Synthetic but Controlled)

3.1 Users

Total users: N_u Types: Honest ($p_{\text{correct}} \approx 0.9$) Noisy ($p_{\text{correct}} \approx 0.5$) Adversarial ($p_{\text{correct}} \approx 0.1$) Bots (coordinated, high rate)

3.2 Posts

Total posts: N_p Labels: True (+1) False (-1) Ambiguous

3.3 Interaction Model

- time-stepped simulation
- stochastic user participation
- controlled coordination groups

4. Experimental Scenarios

Each scenario isolates a failure mode.

Scenario A: Clean Environment

Only honest users

Goal: baseline accuracy and calibration

Scenario B: Noisy Participation

mix of honest + noisy users

Goal: robustness to randomness

Scenario C: Coordinated Attack

bot groups vote synchronously on false posts

Goal: resistance to manipulation

Scenario D: Sybil Attack

many low-history users influence posts

Goal: effectiveness of experience weighting

Scenario E: Location Spoofing (Phase 4)

users provide inconsistent locations

Goal: spatial trust effectiveness

Scenario F: Early Detection

few interactions available

Goal: benefit of ML augmentation

5. Evaluation Metrics

5.1 Credibility Accuracy

Accuracy = $\sum_j \mathbb{1}((C_j) = y_j)$

5.2 Calibration (Brier Score)

$\sum_j (C_j - y_j)^2$

5.3 Time to Correct Classification

time until C_j crosses threshold θ

5.4 Attack Success Rate

false posts incorrectly classified as true

5.5 Alert Precision / Recall

precision = correct alerts / total alerts recall = detected events / total true events

5.6 User Weight Quality

$\text{Corr}(w_i, \text{true reliability}_i)$

5.7 Anomaly Detection Quality

precision / recall for adversarial users

6. Experimental Procedure

Step 1: Initialize

generate users and posts assign ground truth

Step 2: Simulate Interactions

run for T time steps generate events per scenario

Step 3: Run System Variant

apply $M_0 \rightarrow M_5$ separately

Step 4: Collect Metrics

store metrics over time

Step 5: Repeat

multiple runs for statistical confidence

7. Results

7.1 Accuracy Across Models

Model Accuracy M0 ... M1 ... M2 ... M3 ... M4 ... M5 ...

7.2 Attack Resistance

Model Attack Success ↓ M1 high M2 medium M3 low

7.3 Early Detection

Model Time ↓ M1 slow M5 fast

7.4 Alert Quality

Model Precision Recall M3 ... M4 ...

8. Key Findings

9. User Modeling Helps but Is Insufficient

Phase 2 improves accuracy but fails under coordinated attack.

2. Graph Trust Is Critical

Phase 3 significantly reduces:

coordinated attack success

3. Spatial Filtering Improves Relevance

Phase 4 reduces:

false alerts

4. ML Enables Early Detection

Phase 5 improves:

low-data scenarios

9. Ablation Analysis

Remove components individually:

- remove anomaly → performance drops
- remove graph → attack succeeds
- remove spatial → alert noise increases

10. Limitations

Be explicit:

synthetic data may not capture full real-world complexity ML depends on training quality hyperparameters require tuning

11. Reproducibility

Provide:

random seeds parameter values code references

12. Conclusion

The system demonstrates that:

- layered modeling (user + network + spatial + ML) is necessary
- each phase contributes distinct robustness
- coordination-aware trust is the most critical component

13. Visualizations (Required for Demo)

Include:

$C_j(t)$ over time (true vs false posts) attack scenario comparison (M2 vs M3) alert precision-recall curves
anomaly score distributions

14. Final Insight

Your report should make one thing clear:

The system is not one model—it is a hierarchy of defenses